Robust Flood Area Segmentation through Federated Learning on Distributed Remote Sensing Data

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***Abstract*—Floods are among the most devastating and frequent natural disasters, often resulting in severe humanitarian, ecological, and economic consequences. Accurate and timely flood mapping is critical for informed emergency response and disaster risk management. While deep learning has significantly advanced flood area segmentation using remote sensing imagery, traditional methods rely on centralized data aggregation, posing substantial challenges due to privacy restrictions, data heterogeneity, and data ownership concerns. This paper presents a federated learning (FL) framework for robust flood segmentation across distributed, multisource remote sensing data. By enabling multiple institutions to collaboratively train models without exchanging raw data, the proposed system preserves data privacy while leveraging the diversity of geographically and sensor-varied datasets. Experimental results show that our federated approach achieves segmentation performance comparable to centralized training and generalizes better across unseen domains. This framework lays the foundation for scalable, privacy-preserving, and operationally viable flood mapping, especially in data-sensitive environments.**

***Keywords*—flood area; segmentation; detection; deep learning; federated learning**

I. Introduction

Flooding remains among the most frequent and devastating natural disasters worldwide, causing loss of life, large-scale displacement, and economic damage. According to the United Nations Office for Disaster Risk Reduction (UNDRR), the number of major flood events more than doubled between 1980–1999 and 2000–2019, increasing from 1,389 to 3,254 incidents globally [1]. The increasing frequency and severity of floods due to climate change have further emphasized the need for effective disaster management strategies [2]. One of the most crucial components of such strategies is timely and accurate flood area segmentation from remote sensing data, which supports early warning systems, damage assessment, and emergency response.

Remote sensing platforms—including satellites like Sentinel-1 and Sentinel-2, aerial drones, and other earth observation systems- offer an unparalleled ability to observe flood extents over vast and inaccessible regions. Deep learning models, particularly convolutional neural networks (CNNs), have shown promise in automating the extraction of flooded areas from satellite imagery. However, the performance of such models depends heavily on large, annotated datasets that are rarely centralized or uniformly accessible [3]. The need to aggregate data from different institutions often clashes with privacy, security, and sovereignty constraints—especially when dealing with governmental, NGO, or international partners managing sensitive geographic data. Furthermore, flood-related imagery tends to be heterogeneous, originating from different sensor types (e.g., SAR, optical, multispectral) and spanning a wide variety of terrains and climates [4]. Traditional deep learning models trained on localized or biased datasets struggle to generalize to unseen domains. This poses a critical challenge for national-scale or cross-border flood prediction systems.

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To overcome these limitations, we propose a federated learning (FL) framework for flood area segmentation. FL is a decentralized machine learning paradigm where local clients (e.g., regional institutions or agencies) train models on their proprietary datasets and only share model updates, not raw data, with a central aggregator. This process preserves privacy while allowing knowledge transfer across institutions. We hypothesize that the collaborative training across geographically diverse and sensor-varied data domains will lead to more generalizable and robust flood segmentation models.

A. Motivation

Over the past two decades, climate-related disasters have affected over 4.2 billion people worldwide, causing economic losses approaching $3 trillion [1]. In 2022, Pakistan alone experienced unprecedented monsoon floods that submerged one-third of the country, affecting over 33 million people and causing an estimated $30 billion in damages [5, 6]. These figures highlight the urgent need for timely and accurate flood mapping to support disaster response, especially in vulnerable regions of the Global South. However, deploying deep learning models in such contexts is constrained by fragmented data ownership, privacy regulations, and limited access to labeled imagery. With remote sensing platforms increasingly diversified, ranging from public satellites to locally operated drones, there is a pressing need for collaborative solutions that respect data sovereignty. Federated learning offers a promising paradigm by enabling decentralized model training across institutions, turning data heterogeneity into an advantage and enhancing segmentation robustness across varied geographies.

B. Related Work

Flood mapping using remote sensing has seen rapid progress with the application of deep learning, especially in semantic segmentation. U-Net [7] and its variants remain widely used due to their encoder–decoder structure and ability to generalize from limited training data. Recent studies have explored different architectures for flood detection from SAR imagery. In [8], U-Net and Feature Pyramid Network (FPN) were applied to Sentinel-1 data, achieving stable performance in cloudy conditions typical of flood events. The approach in [9] focused on enhanced U-Net variants for water body segmentation, enabling more accurate flood monitoring. A broader comparison in [10] evaluated multiple segmentation models and highlighted the trade-off between accuracy and computational cost for real-world deployment.

Optimized hybrid CNN architectures have been proposed to improve flood detection speed and hardware compatibility. In [11], a model was adapted for real-time inference on satellite platforms, reducing onboard processing delays. Urban flood mapping has also advanced, with [12] linking semantic segmentation outputs to early warning systems to improve response coordination. The U-Net family has continued to evolve, with models such as U-Net++, WU-Net, and others addressing specific segmentation challenges. In [13], a comparative framework was developed for pluvial flood sensitivity analysis in tropical zones, supporting region-specific risk mapping. A deep learning model focused on real-time flood segmentation and emergency detection was presented in [14], targeting smart city integration. In [15], UAV-based flood monitoring was validated using aerial imagery, showing strong potential for rapid, low-altitude assessment in flood-prone areas.

Federated learning (FL), though mature in sectors like mobile computing and healthcare, is still emerging in remote sensing applications [16]. Federated learning has been applied in UAV-based remote sensing to enable distributed model training without centralizing aerial data. One approach integrates fog computing with UAV networks to perform aerial image classification efficiently while minimizing communication and energy costs [17]. Another uses conditional federated learning to support privacy-preserving collaboration among drones for post-disaster infrastructure monitoring, allowing shared model improvement without exposing sensitive visual data [18]. Initial studies point to its potential in flood mapping, where centralized data access is limited. In [19], FL was combined with blockchain to support UAV-based flood detection, ensuring data privacy and system integrity. A related approach in [20] proposed a privacy-preserving FL framework for land area classification, which holds relevance for flood segmentation tasks. These contributions highlight the feasibility of distributed learning methods in geospatial analysis under privacy constraints.

II. Methodology and Experiments

A. Data Collection and Preprocessing

This study utilized 290 aerial drone images and corresponding binary masks of 512×512 from the Flood Area Segmentation dataset available on Kaggle [21]. Each image was in .jpg format, and each mask was in .png format with pixel-level labels indicating flooded (foreground) and non-flooded (background) areas. Images and masks were loaded using a custom preprocessing function. All images were converted to grayscale and normalized by dividing pixel values by 255. Masks were binarized by setting all non-zero pixels to 1 (flood) and zero pixels to 0 (non-flood). The dataset was split into training and validation sets using an 80:20 ratio with a fixed random seed. An analysis of the training masks showed a class imbalance: background pixels (Class 0) made up 60%, while flood pixels (Class 1) accounted for 40%.

B. Federated Data Partitioning

To simulate a realistic federated learning setup, the training data was partitioned across three clients with heterogeneous data distributions. Each client was assigned a non-identical number of samples and varying proportions of flood-affected (foreground) pixels to reflect practical disparities in data volume and flood exposure. The details of data partitioning are given in Table I. This variation enables us to assess the robustness of the federated model under non-IID (non-independent and identically distributed) conditions, which are typical of multi-institutional deployments where regional differences in flooding frequency and sensor coverage result in diverse data characteristics.

TABLE I. Data split across federated clients showing sample count and foreground pixel ratio.

Client	Number of Samples	Foreground Pixel Ratio (%)
Client 1	116	28.25
Client 2	54	51.15
Client 3	62	50.28

C. Model Architecture

We employ a U-Net-based convolutional neural network for semantic segmentation of flood regions [7]. The model takes grayscale input images of size 512×512 and outputs a binary flood mask of the same dimension. It follows an encoder-decoder architecture with four downsampling and upsampling stages. Each downsampling block consists of a convolutional layer followed by max-pooling and dropout, progressively increasing the number of filters. The decoder mirrors this structure using transposed convolutions for upsampling, with skip connections that concatenate encoder features at each level to retain spatial details. A final 1×1 convolution followed by a sigmoid activation generates pixel-wise flood probabilities. Dropout regularization is applied throughout to improve generalization across heterogeneous datasets.

D. Federated Training Setup

Figure 1 illustrates the typical communication and training flow in a federated learning system. Multiple distributed clients (local data holders) collaboratively train a global model. Each client independently trains a local model using its private dataset. Instead of sharing raw data, clients send only their model updates (e.g., gradients or trained weights) to a central server. The server aggregates these updates to create an improved global model, which is then distributed back to the clients for subsequent training rounds. This iterative

process enables robust global model learning from diverse, decentralized data while ensuring the privacy of individual client datasets.

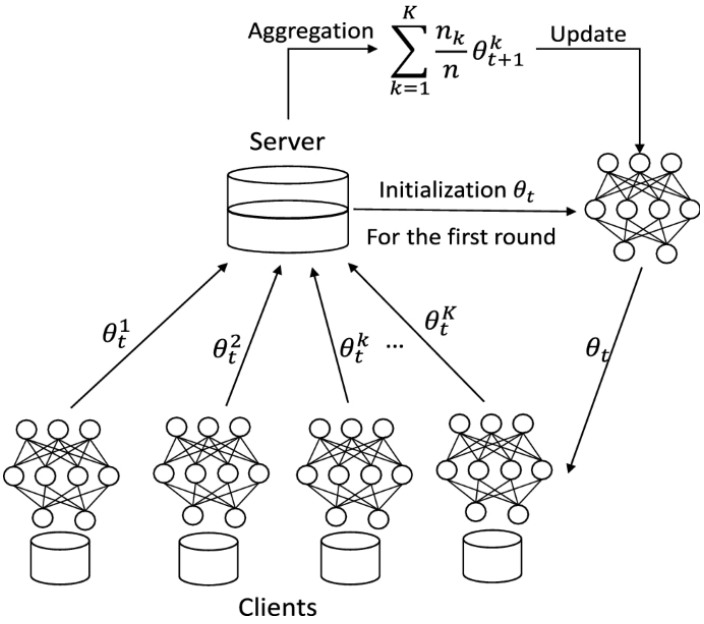


Fig. 1. Standard Federated Learning Architecture. (Adapted from [22])

E. Training Procedure

For the centralized setup, a U-Net-based model was trained using a standard supervised learning pipeline. The training data, composed of labeled flood imagery, was split into training and validation sets. The model was optimized using the Adam optimizer with a learning rate of 0.001 and binary cross-entropy as the loss function. To evaluate segmentation performance during training, a custom Intersection-over-Union (IoU) metric was monitored. The model was trained for 300 epochs with a batch size of 32, and the best weights were saved using a model checkpoint based on validation IoU.

In the federated setup, a synchronous federated averaging (FedAvg) scheme was employed across three clients, each representing a different local dataset with varying flood foreground ratios. At each communication round, each client received the current global model weights, trained a local copy for one epoch on their private data, and returned the updated weights. The server aggregated these updates using weighted averaging based on the number of samples per client. The aggregated weights were then used to update the global model. This process was repeated for 300 rounds. The best global model weights were selected based on the maximum validation IoU observed across all rounds.

III. Results and Discussion

A. Quantitative analysis

This section presents a detailed quantitative comparison between the **centralized** and **federated learning (FL)** models for flood area segmentation. Both models were evaluated using standard semantic segmentation metrics, including Precision, Recall, F1-score, Intersection over Union (IoU), Pixel Accuracy (PA), Mean Pixel Accuracy (MPA), and Frequency Weighted Intersection over Union (FWIoU).

TABLE II. Quantitative comparison between centralized and federated models.

Metric	Centralized Model	Federated Model
Precision	0.8443	0.7900
Recall	0.8638	0.8883
F1-score	0.8539	0.8363
Intersection over Union (IoU)	0.7450	0.7186
Mean Pixel Accuracy (MPA)	0.8850	0.8691
Frequency Weighted IoU (FWIoU)	0.780	0.7645

The centralized model, which was trained with pooled data, achieved the highest performance across most metrics. It yielded a precision of 0.8443, a recall of 0.8638, an F1-score of 0.8539, and an IoU of 0.7450. The high mean pixel accuracy (MPA) of 88.5% confirms its reliable class-wise segmentation performance. A summary of the results is presented in Table II.

In contrast, the federated model performed competitively, despite the decentralized and heterogeneous data distribution across clients. It demonstrated a higher recall (0.8883), indicating stronger sensitivity to true flood areas, critical in disaster response applications. However, this came at a slight cost to precision (0.7900), implying a modest increase in false positives. The F1-score of 0.8363 and IoU of 0.7186 validate the model's effectiveness in real-world settings. Moreover, the FL model achieved a pixel accuracy (PA) of 86.49%, MPA of 86.91%, and frequency-weighted IoU of 0.7645—further confirming its robustness across diverse geographies and sensor types.

Although centralized training yields slightly better overall performance, the federated model offers a compelling alternative where data privacy, sovereignty, or communication constraints prevent centralized data aggregation. The relatively small performance gap (typically within 1–3%) demonstrates that federated learning can serve as a viable and scalable solution for flood mapping in decentralized, multi-institutional environments.

B. Qualitative Assessment

Figure 2 presents a visual comparison of flood area segmentation outputs for several representative scenes. The figure includes: (a) original grayscale aerial imagery, (b) ground truth binary flood masks, (c) segmentation results produced by the *federated model*, and (d) outputs from the *centralized model*. Both models generally align well with the ground truth, correctly identifying major flood-affected regions. However, several differences are evident upon closer inspection.

The centralized model produces cleaner and more sharply delineated segment boundaries, particularly in complex urban scenes. These outputs reflect its higher precision (0.8443) and F1-score (0.8539), indicating a better balance between false positives and false negatives. In contrast, the federated model demonstrates a tendency toward over-segmentation, labeling some non-flood areas as flooded as visible in the figure. This behavior aligns with its higher recall (0.8883) and lower precision (0.7900), suggesting the model favors sensitivity,

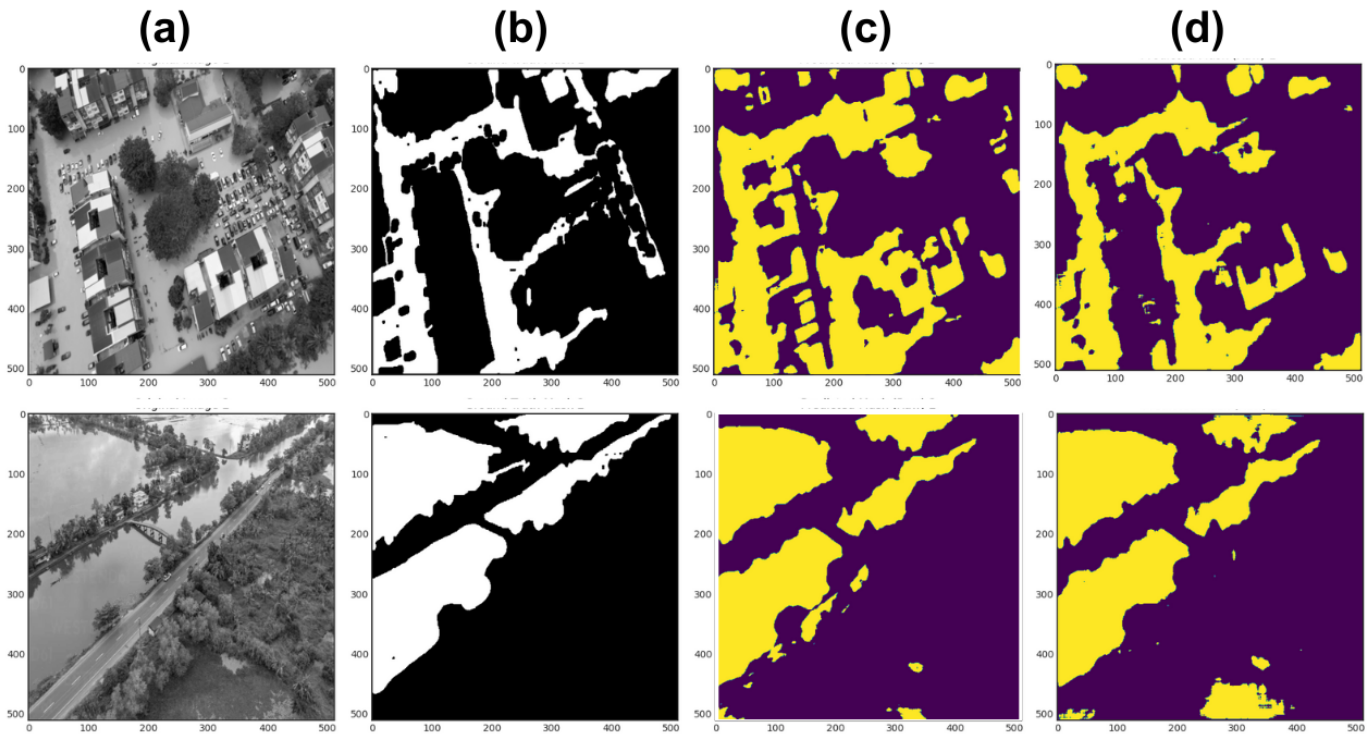


Fig. 2. Qualitative comparison of flood segmentation: (a) input image, (b) ground truth, (c) federated model prediction, (d) centralized model prediction. Yellow: predicted flood, Purple: non-flood.

which is advantageous in scenarios where underestimating flood extent carries high risk.

Despite the lack of centralized data aggregation, the federated model demonstrates strong generalization across diverse landscapes and imaging conditions. The qualitative evidence supports the quantitative findings and validates that the federated learning approach can achieve segmentation results close to the centralized upper bound while preserving data privacy and sovereignty.

IV. Conclusion

Floods cause severe damage to humans, the environment, and the economy. Timely flood mapping is essential for effective disaster response. While deep learning has improved flood segmentation, centralized training faces challenges from privacy limits, data heterogeneity, and ownership constraints. This study evaluated a federated learning framework for flood area segmentation using remote sensing imagery. The results demonstrate that federated models can achieve performance levels close to those of centralized training, with only a minor trade-off in precision and overall accuracy. Despite the decentralized nature of data and the presence of heterogeneity, the federated approach showed strong generalization and maintained high recall, which is critical in disaster response scenarios. Both quantitative metrics and qualitative visualizations confirm the practicality of federated learning for flood mapping, where data privacy, institutional boundaries, or limited connectivity restrict central aggregation.

Future work will focus on extending this framework to support multi-class flood impact assessment, including infrastructure damage and water depth estimation. Adaptive federated optimization strategies and client selection techniques will be explored to further reduce performance gaps. Integration with real-time data streams, such as UAV feeds and IoT sensors, will also be investigated to support dynamic flood monitoring and alert systems. In addition, the framework will be tested across other environmental monitoring tasks, such as landslide and wildfire detection, to evaluate its broader applicability.

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